



CAMBRIDGE IGCSE

3 Coordinate geometry

0580 PAPER 4 - CALCULATOR

QUESTIONS + MARK SCHEME ANSWERS

34 complete questions

SUBTOPIC

Gradients and equations of lines

Each question is followed immediately by its official mark scheme answer.

	2	(a)	1.62 or 1.62...	1		
		(b) (i)	7	1		
		(ii)	4	1		
		(iii)	7	1		
		(iv)	1	1		

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(c) (i)	8π final answer	1	2
	(ii) 72π final answer	2FT	FT <i>their</i> (c)(i) $\times 9$ in terms of π M1 for area factor of 3^2 or 9 or [new a] = 12, [new b] = 6

7(a)	$x = 0$	1	
7(b)	Tangent ruled at $x = 0.5$	B1	No daylight between tangent and curve at point of contact
	–9 to –6.5	2	dep on ruled tangent or close attempt at tangent at $x = 0.5$ M1 for rise/run also dep on tangent or close attempt at tangent at $x = 0.5$
7(c)(i)	0 2.4 or better 4	3	B1 for each
7(c)(ii)	Correct smooth curve	4	B3FT for 6 or 7 correct plots or B2 FT for 4 or 5 correct plots or B1 FT for 2 or 3 correct plots FT <i>their</i> table
7(d)	$x^3 + 3x + 4 = 10 - 8x^2$ and correctly completed	1	
7(e)	line $y = -2x + 2$ drawn and –0.45 to –0.35 nfww	3	B2 for ruled $y = -2x + 2$ or B1 for $-2x + 2$ seen or for line $y = -2x + c$ drawn or for $y = cx + 2$ ($c \neq 0$) drawn and B1 for –0.45 to –0.35 nfww

7(a)	$[y =] 4x + 5$	3	B2 for answer $[y =] 4x + c$ oe (c can be numeric or algebraic) OR M2 for $\frac{y-9}{x-1} = \frac{9-(-3)}{1-(-2)}$ oe OR M1 for $\frac{9-(-3)}{1-(-2)}$ oe or for M1 for correct substitution of $(-2, -3)$ or $(1, 9)$ into $y = (\text{their } m)x + c$ oe
7(b)	76[.0] or 75.96...	2	M1 for $\tan[] = 4$ oe

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				M1 for $-\frac{1}{\text{their gradient from (a)}}$ soi M1 for correct substitution of (3.5, 2) into $y = (\text{their } m)x + c$ oe	
	7(c)(ii)	(−4.5, 4)	2	B1 for each value or for $\begin{pmatrix} -8 \\ 2 \end{pmatrix}$ seen	

10(a)(i)	15.7 or 15.65...	3	M2 for $\sqrt{(4-10)^2 + (4-3)^2}$ oe or M1 for $(-4-10)^2 + (4-3)^2$ oe
10(a)(ii)	$\frac{-10-4}{4-3} [= -2]$ oe	M1	
	$10 = -2(-3) + c$ Or $-4 = -2(4) + c$ and correct completion to $y = -2x + 4$	A1	
10(a)(iii)	$y = \frac{1}{2}x + \frac{11}{4}$ oe	4	M1 for grad = $\frac{1}{2}$ soi M1 for [midpoint =] $(\frac{1}{2}, 3)$ M1 for substitution of $(\frac{1}{2}, 3)$ into <i>their</i> $y = mx + c$ oe
10(b)(i)	$\left(-\frac{1}{3}, -\frac{22}{27}\right)$ oe and $(-5, 50)$	6	B2 for $3x^2 + 16x + 5$ Or B1 for one correct M1 for derivative = 0 or <i>their</i> derivative = 0 M1 for $[x =] -\frac{1}{3}$ and $[x =] -5$ B1 for $-\frac{22}{27}$ and 50
10(b)(ii)	$\left(-\frac{1}{3}, -\frac{22}{27}\right)$ minimum $(-5, 50)$ maximum with correct reasons	3	B2 for one correct with reason or M1 for correct attempt e.g. 2 nd derivatives, gradients or sketching

SUBTOPIC

Coordinates, length and midpoint

Each question is followed immediately by its official mark scheme answer.

10 (a) (b)	b – a or – a + b $\frac{4}{5}\mathbf{b} - \frac{3}{10}\mathbf{a}$ or $\frac{1}{10}(8\mathbf{b} - 3\mathbf{a})$	1 4	B3 for correct unsimplified expression in a and b or M1 for $\vec{XA} + \vec{AC} + \vec{CM}$ or $\vec{XB} + \vec{BM}$ or $-\frac{1}{5}(\text{their } \mathbf{a}) + \mathbf{b} - \frac{1}{2}\mathbf{a}$ or $\frac{4}{5}(\text{their } \mathbf{a}) + \frac{1}{2}\mathbf{a}$ and M1 indep for $\pm\frac{1}{5}$ oe or $\pm\frac{4}{5}$ oe used After zero scored, SC2 for answer $\frac{1}{4}(3\mathbf{b} - \mathbf{a})$ or $\frac{3}{4}\mathbf{b} - \frac{1}{4}\mathbf{a}$	
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11(a)(i)	$8\mathbf{b} - 4\mathbf{a}$ oe	1	
11(a)(ii)	$6\mathbf{b}$	1	
11(a)(iii)	$6\mathbf{b} - 2\mathbf{a}$ or $2(3\mathbf{b} - \mathbf{a})$	1	FT $-2\mathbf{a} + \text{their (a)(ii)}$
11(b)	$2 : 1$ oe final answer	3	Dep on correct $\overrightarrow{BC}$ or correct $\overrightarrow{AC}$ seen B2 for $\overrightarrow{BC} = 4\mathbf{b} - 2\mathbf{a}$ or M1 for a correct route for $\overrightarrow{BC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ or for a correct route for $\overrightarrow{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ If no/incorrect working seen then SC1 for final answer of $2 : 1$ (oe)

Page 11 of 12		$\pi(5x)^2$ or $\pi(5x)(13x)$	M1	Accept $25\pi x^2$
		$\pi(5x)^2 + \pi(5x)(13x) = 4\pi x^2$	M1	
		$r^2 = \frac{90\pi}{4\pi}x^2 = \frac{45}{2}x^2$	A1	With all steps shown and no errors seen
Question	Answer	Marks	Partial Marks	
11(a)	(0, 16) (4, -16)	6	M1 for $3x^2$ or $12x$ A1 correct $3x^2 - 12x$ B1 setting <i>their</i> $dy/dx = 0$ M1 for factorising <i>their</i> dy/dx A1 $x = 0$ and $x = 4$ A1 (0, 16) and (4, -16)	
11(b)	(0, 16) maximum with correct reason (4, -16) minimum with correct reason	3	B2 for both correct with no/one reason or B1 for one correct (with no reasons) or M1 correct attempt to find e.g. second derivative or gradients	

9(a)(ii)	$\begin{pmatrix} -10 \\ 15 \end{pmatrix}$	1	
9(a)(iii)	3.61 or 3.605 to 3.606	2	M1 for $(-2)^2 + 3^2$ oe
9(b)(i)(a)	$\frac{1}{2} \mathbf{c}$	1	
9(b)(i)(b)	$\mathbf{a} + \frac{1}{2} \mathbf{c}$ oe	1	FT $\mathbf{a} + \textit{their (b)(i)(a)}$

9(b)(ii)(a)	$\overrightarrow{OP} = \frac{1}{3}(2\mathbf{a} + \mathbf{c})$ oe and $\overrightarrow{OQ} = \frac{1}{2}(2\mathbf{a} + \mathbf{c})$ oe OR $\overrightarrow{OP} = \frac{2}{3}(\mathbf{a} + \frac{1}{2}\mathbf{c})$ OR $\overrightarrow{PQ} = \frac{1}{3}(\mathbf{a} + \frac{1}{2}\mathbf{c})$ and correct comment e.g. have the same base vector or that they are multiples of one another and they share a common point OR e.g. $\overrightarrow{OQ} = 1.5\overrightarrow{OP}$, $2\overrightarrow{PQ} = \overrightarrow{OP}$	2	B1 for $\overrightarrow{OP}$ or $\overrightarrow{PQ}$ factorised or for correct multiplicative statement on relationship without factorised vectors e.g. $\overrightarrow{OQ} = 1.5\overrightarrow{OP}$, $\frac{2}{3}\overrightarrow{OQ} = \overrightarrow{OP}$, $2\overrightarrow{PQ} = \overrightarrow{OP}$, $1.5\left(\frac{2}{3}\mathbf{a} + \frac{1}{3}\mathbf{c}\right) = \mathbf{a} + \frac{1}{2}\mathbf{c}$
9(b)(ii)(b)	1.5 oe	1	
10(a)	(2, -10) and (-2, 22)	5	B2 for $3x^2 - 12$ isw or B1 for $3x^2 + k$ or $px^2 - 12$ ($p \neq 0$) or for $3x^2 - 12 + 6$ isw M1 for setting <i>their</i> derivative = 0 or $\frac{dy}{dx} = 0$ B1 for $x = \pm 2$ or for one correct coordinate pair
10(b)	(2, -10) minimum with correct reason or sketch (-2, 22) maximum with correct reason or sketch	3	B2 for 1 correct with correct reasoning or B2FT for correct evaluation with correct 2nd derivative for both of <i>their</i> different x values or M1 for showing [2nd derivative =] $6x$ or gradients for one value on either side of one correct stationary point or for reasonable sketch of cubic

6(a)(i)	$\begin{pmatrix} -3 \\ 3 \end{pmatrix}$	1	
6(a)(ii)	$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$	1	
6(a)(iii)	3.61 or 3.605 to 3.606	2	M1 for $2^2 + 3^2$ oe
6(b)	(6, 1)	2	B1 for each

	6(c)	23	4	R3 for correct unsimplified expression for	
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5(a)(i)	2[.00] or 2.002 to 2.003 nfw	3	M2 for $\sqrt{4.8^2 + 5.6^2 - 2 \times 4.8 \times 5.6 \times \cos 20.4}$ OR M1 for $4.8^2 + 5.6^2 - 2 \times 4.8 \times 5.6 \times \cos 20.4$ A1 for 4.01[17...] or 4.012
5(a)(ii)	4.1[0] or 4.11 or 4.100 to 4.107 cao	2	M1 for $\tan 64 = \frac{AX}{\text{their (a)(i)}}$ or for $\frac{AX}{\sin 64} = \frac{\text{their (a)(i)}}{\sin(90 - 64)}$ oe
5(a)(iii)	6.96	2	M1 for $\frac{1}{2} \times 4.8 \times 2.9$ oe

	5(b)	11.3 or 11.31..	5	M4 for $2 \times \frac{8}{\sin(45)} \times \sin 30$ or B4 for $PM = 5.65[685\dots]$ or 5.66 or better OR B1 for <i>angle RPM</i> = 45° M2 for $\frac{8}{\sin(\text{their } 45)} \times \sin 30$ or M1 for implicit form	
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10(a)(i)	(15, 6)	2	B1 for each
10(a)(ii)	$\begin{pmatrix} 3 \\ 24 \end{pmatrix}$	1	
10(a)(iii)	13.6 or 13.60...	2	M1 for $(-11)^2 + 8^2$ oe
10(b)(i)	$\mathbf{a} + \frac{3}{5}(\mathbf{b} - \mathbf{a})$ or $\mathbf{b} + \frac{2}{5}(\mathbf{a} - \mathbf{b})$ leading to $\frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}$ with no errors	M3	M2 for $[\overrightarrow{MR} =] \frac{3}{5}(\mathbf{b} - \mathbf{a})$ oe or $[\overrightarrow{NR} =] \frac{2}{5}(\mathbf{a} - \mathbf{b})$ oe or M1 for $\overrightarrow{MN} = \mathbf{b} - \mathbf{a}$ or $\overrightarrow{NM} = \mathbf{a} - \mathbf{b}$ or a correct route for $\overrightarrow{OR}$
10(b)(ii)(a)	$k = 5, c = 10$	4	B2 for $c = 10$ or M1 for $c(\frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}) = \mathbf{b} + 4\mathbf{a} + k\mathbf{b}$ oe or for $\frac{2}{5}c = 4$ and M1 for $\frac{3}{5} \times \text{their } c = k + 1$
10(b)(ii)(b)	$3\mathbf{a} + 6\mathbf{b}$ final answer	1	FT $3\mathbf{a} + (\text{their } k + 1)\mathbf{b}$

6(a)	$\begin{pmatrix} 4 \\ -3 \end{pmatrix}$	2	B1 for $\begin{pmatrix} 6 \\ -10 \end{pmatrix}$ or answer $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$
6(b)(i)	$(-4, -1)$	1	
6(b)(ii)	7.21 or 7.211...	2	M1 for $(-6)^2 + 4^2$
6(c)(i)(a)	$2\mathbf{c} - \mathbf{a}$	1	
6(c)(i)(b)	$\mathbf{c} - \mathbf{a}$	1	
6(c)(i)(c)	$\frac{1}{5}(\mathbf{a} + 8\mathbf{c})$ final answer	2	M1 for $[\overrightarrow{AP} =] \frac{4}{5} \times \text{their}(2\mathbf{c} - \mathbf{a})$ or $[\overrightarrow{BP} =] \frac{1}{5} \times - \text{their}(2\mathbf{c} - \mathbf{a})$ or for a correct vector route using the lines on the diagram
6(c)(i)(d)	$\frac{4}{5}(-\mathbf{a} + \mathbf{c})$ final answer	2	M1 for $[\overrightarrow{QP} =] -\frac{4}{5}\mathbf{c} + \frac{4}{5} \times \text{their}(2\mathbf{c} - \mathbf{a})$ or for a correct vector route
6(c)(ii)	$[\overrightarrow{QP} \text{ is}] \text{ parallel } [\text{to } \overrightarrow{CB}]$ $\overrightarrow{QP} = \frac{4}{5} \overrightarrow{CB}$ oe	2	Dep both statements consistent with <i>their (c)(i)(b)</i> and <i>their (c)(i)(d)</i> and both vectors in terms of $\mathbf{a}$ and $\mathbf{c}$ B1 for each dep on statement consistent with <i>their (c)(i)(b)</i> and <i>their (c)(i)(d)</i> and both vectors in terms of $\mathbf{a}$ and $\mathbf{c}$

SUBTOPIC

Parallel and perpendicular lines

Each question is followed immediately by its official mark scheme answer.

	8	(a)	$(4, 6)$	1, 1	
		(b)	4.47 or 4.472	3	M2 for $\sqrt{(8-4)^2 + (5-3)^2}$ or better or M1 for $(8-4)^2 + (5-3)^2$ or better
		(c)	$y = 2x - 2$ oe	3	B2 for $2x - 2$ or $y = 2x + c$ oe or M1 for $[m =] \frac{8-4}{5-3}$ oe soi by $2x$ and M1 for $(3, 4)$ or $(5, 8)$ or <i>their</i> midpoint substituted into <i>their</i> $y = mx + c$ with m numerical
		(d)	-3	3	M1 for use of gradient $\times$ <i>their</i> $m = -1$ soi by $-\frac{1}{2}$ M1 for $r = \text{their gradient} \times 6$ [+0]

9 (a)	19[.0] or 18.97.. nfw	3	M2 for $\sqrt{(4 - -2)^2 + (13 - -5)^2}$ oe or M1 for $(4 - -2)^2 + (13 - -5)^2$ oe
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				or $y = \{their\ gradient\ from\ (b)\}x + c$ If 0 scored, SC1 for answer $3x - 5$ 5 B2FT for $-\frac{1}{3}x + c$ (c can be numeric or algebraic) FT $-1/$ <i>their</i> gradient from (b) or M1 for $-1/$ <i>their</i> gradient from (b) soi and B1 for [midpoint of $AB =$] (1, 4) and M1 for substitution of (1, k) or (k, 4) into a linear equation	
	(d)	$y = -\frac{1}{3}x + \frac{13}{3}$ oe isw			

MARK SCHEME ANSWER

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	7(a)	$(-0.5, 3)$	2	B1 for one correct value
	7(b)	$[y =] -2x + 2$ final answer	3	M1 for $\frac{-2-8}{2-3}$ or better M1 for substitution of $(-3, 8)$ or $(2, -2)$ or <i>their</i> midpoint into $y = mx + c$ with <i>their</i> m
	7(c)	$y = -2x + 7$ oe	2FT	FT <i>their</i> (b) M1 for $y = (their-2)x + k$ ($k \neq 2$) or $y = kx + 7$ ($k \neq 0$) If zero scored, SC1 for $(their - 2)x + 7$
	7(d)	$x - 2y + 9 = 0$ or $2y - x - 9 = 0$ oe	4	B3 for any correct equivalent in wrong form Or M2 for $y = \frac{1}{2}x + k$ oe (FT negative reciprocal of <i>their</i> gradient in (b)) or M1 for $grad = \frac{1}{2}$ (FT negative reciprocal of <i>their</i> gradient in (b)) M1 for substitution of $(1, 5)$ into $y = mx + c$ oe with <i>their</i> m

	$b = -\frac{5}{8}$ or $-6\frac{5}{8}$ or -6.125		or for $2x^2 + 4ax + 2a^2 + b$
8(a)(i)	5	1	
8(a)(ii)	$-\frac{3}{2}$ oe	1	
8(b)	$\left(\frac{4}{5}, 0\right)$ oe	2	M1 for $5x - 4 = 0$ soi

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	8(e)	13	3	M2 for $(4 + 9) \times \textit{their } 2 \div 2$ oe or B1 for 9 oe or 4 or -4 seen	
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	10(a)	10.8 or 10.81 to 10.82	3	M2 for $\sqrt{(6-3)^2 + (-2-4)^2}$ oe or M1 for $(6-3)^2 + (-2-4)^2$ oe
	10(b)(i)	(6, 4)	2	B1 for each
	10(b)(ii)	2	2	M1 for $\frac{12 - (-4)}{10 - 2}$ oe
	10(b)(iii)	$y = -\frac{1}{2}x + 4$ oe final answer	3	M1 for gradient = $-\frac{1}{2}$ or $-\frac{1}{\text{their (b)(ii)}}$ M1 for (2, 3) substituted into <i>their</i> $y = mx + c$ or $y - y_1 = m(x - x_1)$ oe

8(a)(i)	$\times$ $\checkmark$ $\checkmark$ $\times$ $\times$ $\checkmark$	4	B3 for 5 correct B2 for 4 correct B1 for 3 correct
8(a)(ii)	$\begin{pmatrix} 5 \\ 3 \end{pmatrix}$	1	Fraction line and/or missing brackets scores 0
8(a)(iii)	$\begin{pmatrix} 4 & 8 \\ 1 & 2 \end{pmatrix}$	2	B1 for 2 or 3 correct elements (dep on 2×2 matrix)
8(a)(iv)	$\frac{1}{2} \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$ oe isw	2	B1 for $k \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$ or determinant = 2 soi
8(b)	Rotation Origin oe 90 [anticlockwise] oe	3	B1 for each
9(a)	$y = -2x + 5$ oe	3	B2 for $-2x + 5$ or M1 for gradient = $-1 \div \frac{1}{2}$ or better M1 for substituting (1, 3) into $y = (their\ m)x + c$ oe If 0 scored SC1 for (1, 3) satisfying their wrong equation ($c \neq 0$) with gradient $\neq \frac{1}{2}$
9(b)(i)	$x \geq 2$ oe $y \leq 5$ oe $y \geq \frac{1}{2}x$ oe	4	SC3 for $x > 2$ and $y < 5$ and $y > \frac{1}{2}x$ OR B1 for $x \geq 2$ B1 for $y \leq 5$ B2 for $y \geq \frac{1}{2}x$ or M1 for $y \geq kx$ ($k > 0$) OR SC2 for all three boundary lines identified but with incorrect sign(s) If 0 scored SC1 for one or two correct boundary lines with incorrect sign(s)

8(a)	(5, 6)	1	
8(b)	$[y =] -\frac{4}{5}x + 3$ nfww	3	B2 for $[y =] -\frac{4}{5}x + c$ nfww or M1 for $\frac{\text{rise}}{\text{run}}$ using any two of $(-5, 7)$ $(0, 3)$ and $(5, -1)$ and B1 for $[y =]mx + 3$ ($m \neq 0$)

				their gradient M1 for (8, 14) substituted into <i>their</i> $y = mx + c$ or $\frac{y-14}{x-8} = m$ or better	
	8(d)(ii)	8.54 or 8.544...	3	M2 for $(14 - \text{their } 6)^2 + (8 - \text{their } 5)^2$ or better or M1 for $14 - \text{their } 6$ and $8 - \text{their } 5$ seen	
	8(d)(iii)	(4, 6)	2	B1 for each	

	3(b)(iv)	4	1	
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	9(a)(i)	$\frac{3}{8}$	2	M1 for $8y = 3x + 20$ or better
	9(a)(ii)	(0, 2.5) oe	1	
	(b)(i)	15.6 or 15.62...	3	M2 for $\sqrt{(9-3)^2 + (-2-8)^2}$ oe seen or M1 for $(9-3)^2$ or $(-2-8)^2$ oe seen
	9(b)(ii)	$y = -\frac{5}{6}x + 4$ oe	3	M1 for gradient $\frac{-2-8}{9-3}$ oe M1 for substituting (6, -1) into a linear equation oe
	9(b)(iii)	$y = \frac{6}{5}x - \frac{3}{5}$ oe	4	M1 for gradient -1 / <i>their</i> $\left(-\frac{5}{6}\right)$ B1 for midpoint at (3, 3) M1 for <i>their</i> midpoint substituted into $y = \text{their } m \times x + c$ oe

11(d)	12	1	
12(a)	17	3	M2 for $3 \times 2x^2 - 7$ or better isw or M1 for $3 \times 2x^2$ oe or $kx^2 - 7$ seen

			into $y = \text{their } px + c$ oe where $\text{their } p \neq 0$
12(b)(iii)	(5, 0)	4	B3 for $\overrightarrow{AD} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$ or $\overrightarrow{DA} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ or coordinates of C $(-7, 6)$ and $\left[\overrightarrow{CD} = \right] \begin{pmatrix} 12 \\ -6 \end{pmatrix}$ oe seen or B2 for $a = b = 2$ soi or coordinates of C $(-7, 6)$ or M1 for $a = b$ oe soi or for $a^2 + b^2 = (\sqrt{8})^2$ oe or $\cos 45 = \frac{a}{\sqrt{8}}$ oe or for $\left[\overrightarrow{DC} = \right] \begin{pmatrix} -12 \\ 6 \end{pmatrix}$ or $\left[\overrightarrow{CD} = \right] \begin{pmatrix} 12 \\ -6 \end{pmatrix}$ seen or $\frac{y-8}{x-5} = 1$ oe or $\frac{y-2}{x-7} = 1$

	9(a)(i)	(3, 1)	1	
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	3(a)	8.25 or 8.246...	3	M2 for $(3--5)^2 + (2-4)^2$ oe or better or M1 for $(3--5)$ and $(2-4)$ oe seen	
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8(a)(i)	$(-0.5, 1)$	2	B1 for each
8(a)(ii)	$\begin{pmatrix} 7 \\ -3 \end{pmatrix}$	2	B1 for each
8(a)(iii)	7.62 or 7.615 to 7.616	2	FT <i>their</i> (a)(ii) M1 for $(\text{their } 7)^2 + (\text{their } -3)^2$ oe
8(a)(iv)	$[y =] -4x - 1$ final answer	3	B2 for answer $-4x + c$ [oe] or for correct equation in different form or for $-4x + -1$ or for $-4m - 1$ OR M1 for $\frac{-5-7}{1-2}$ oe M1 for correct substitution shown of $(-2, 7)$ or $(1, -5)$ or <i>their</i> $(-0.5, 1)$ into $y = (\text{their } m)x + c$ oe OR M1 for $7 = -2m + c$ and $-5 = m + c$ A1 for $m = -4$ and $c = -1$
8(a)(v)	$[y =] \frac{1}{4}x + \frac{11}{4}$ final answer	3	M1 for $\text{grad} = \frac{1}{4}$ oe nfwf soi, FT negative reciprocal of <i>their</i> gradient from (iv) M1 for correct substitution shown of $(5, 4)$ into $y = (\text{their } m)x + c$ oe or, if no substitution shown, $(5, 4)$ satisfies <i>their</i> final linear equation.

	8(b)	$2x^2 + 11x - 21 [= 0]$	M2	or M1 for $8 - 5x = 2x^2 + 6x - 13$ oe or better
		$(2x - 3)(x + 7) [= 0]$ oe or $\frac{-11 \pm \sqrt{11^2 - 4 \times 2 \times (-21)}}{2 \times 2}$ or $-\frac{11}{4} \pm \sqrt{\frac{21}{2} + \left(\frac{11}{4}\right)^2}$ oe	M2	Allow correct method to solve <i>their</i> quadratic equation e.g. formula, complete the square but not for $2x^2 + 6x - 13$ M1 FT <i>their</i> equation for $2x(x + 7) - 3(x + 7) [= 0]$ or $x(2x - 3) + 7(2x - 3) [= 0]$ or $(2x + a)(x + b) [= 0]$ where $ab = -21$ or $2b + a = 11$ OR M1 for $\sqrt{11^2 - 4 \times 2 \times -21}$ or for $\frac{-11 + \sqrt{k}}{2 \times 2}$ or $\frac{-11 - \sqrt{k}}{2 \times 2}$ OR M1 for $\left(x + \frac{11}{4}\right)^2$
		$\left(\frac{3}{2}, \frac{1}{2}\right)$ and $(-7, 43)$	B2	B1 for one correct pair or for 2 correct x-values or 2 correct y-values

8(a)	(22, 11)	2	B1 for each value
8(b)	$\frac{their11-3}{their22-2}$ oe or better	M1	
	$-\frac{1}{their\ m}$	M1	
	Substitution of (12, 7) into $y = (their\ m)x + c$	M1	Accept $y - 7 = their\ m(x - 12)$ oe
	leading to $2y + 5x = 74$ final answer	A1	Without error or omission
8(c)	32	1	
8(d)	145	2	M1 for $\frac{1}{2} \times (their\ 32 - 3) \times 10$ oe or $\frac{1}{2} \times \sqrt{(7-3)^2 + (12-2)^2} \times \sqrt{(their32-7)^2 + (2-12)^2}$ oe

5(a)	$6p^4 - 13p^2 + 6$ final answer	2	B1 for three of $6p^4 - 9p^2 - 4p^2 + 6$ seen
5(b)(i)	175	2	M1 for $\frac{1}{2}(20+30) \times 7$ oe
5(b)(ii)	$\frac{2s-ut}{t}$ or $\frac{2s}{t} - u$ final answer	3	B2 for correct answer but unsimplified e.g. $\frac{s \div t}{0.5} - u$, $\frac{s}{\frac{1}{2}t} - u$, $\frac{s}{0.5t} - u$ OR M1 for correct multiplication by 2 or division by 0.5 M1 for correctly rearranging terms to isolate term in v M1 for correct division by t Max 2 marks if final answer incorrect
5(c)(i)	$(2q-3)(t+2)$ final answer	2	B1 for $t(2q-3) + 2(2q-3)$ or $2q(t+2) - 3(t+2)$

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				or $y = kx + 4$	
6(c)(i)	Gradient = $\frac{-1}{\text{their gradient in } (b)}$		M1	Accept e.g. $2 \times -\frac{1}{2} = -1$ oe or states negative reciprocal of $-\frac{1}{2} = 2$	
	Substituting (2, 3) in <i>their</i> equation.		M1	$3 = 2 \times \text{their } m + c$	
	leading to $y = 2x - 1$		A1	No errors or omissions	
6(c)(ii)	3.35 or 3.354...		5	B2 for $\left(\frac{1}{2}, 0\right)$ soi or x -coordinate of $D = \frac{1}{2}$ or M1 for $2x - 1 = 0$ M2 for $(2 - \text{their } \frac{1}{2})^2 + (3 - \text{their } 0)^2$ oe or M1 for $(2 - \text{their } \frac{1}{2})$ and $(3 - \text{their } 0)$ oe	
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	6(a)	(5, 2) (2, - 2)	4	B3 for 3 correct values or answers for <i>C</i> and <i>D</i> reversed or correct coordinates given on diagram wrongly labelled or B2 for one correct coordinate pair correctly labelled or M2 for <i>A</i> , <i>B</i> , <i>C</i> and <i>D</i> correctly plotted or M1 for <i>A</i> and <i>B</i> correctly plotted If 0 or 1 scored instead award SC2 for answers (-3, 8) and (-6, 4) or answers (1.5, 1.5) and (-2.5, 4.5)
	6(b)(i)	(2.5, 3.5) oe	2	B1 for each
	6(b)(ii)	7.07 or 7.071...	3	M2 for $(6 - -1)^2 + (4 - 3)^2$ oe or M1 for $(6 - -1)$ or $(4 - 3)$ oe
	6(b)(iii)	1	2	$4 - 3$

	11(a)	10	3	M2 for $(1 - -7)^2 + (4 - -2)^2$ oe or M1 for $(1 - -7)$ or $(4 - -2)$ oe
	11(b)	$\frac{4}{3}$ or $\frac{8}{6}$	2	M1 for $\frac{1 - -7}{4 - -2}$ oe
	11(c)	$y = -\frac{3}{4}x - \frac{9}{4}$ or $4y + 3x + 9 = 0$ oe final answers	4	B3 for $-\frac{3}{4}x - \frac{9}{4}$ OR B1 for midpoint $(1, -3)$ M1 for gradient $-\frac{3}{4}$ or $-\frac{1}{\text{their } (b)}$ M1 for substituting <i>their</i> $(1, -3)$ into $y = (\text{their } m)x + c$ or for $\text{their } m = \frac{y - -3}{x - 1}$ oe

12(a)(i)	$\begin{pmatrix} 2 \\ 5 \end{pmatrix}$	1	
12(a)(ii)	$\begin{pmatrix} -6 \\ 4 \end{pmatrix}$	1	
12(b)	$[y =] -\frac{2}{3}x + \frac{19}{3}$ oe	3	M1 for gradient = $\frac{1-5}{8-2}$ oe M1 for substituting (8, 1) or (2, 5) into $y = \text{their } mx + c$
12(c)	$[y =] \frac{3}{2}x - \frac{9}{2}$ oe	4	B1 for (5, 3) oe M1 for gradient = $-\frac{1}{\text{their gradient of } AB}$ M1 substituting <i>their</i> midpoint into $y = \text{their } mx + c$
12(d)	$\frac{65}{6}$ oe	2	M1 for <i>their</i> $\frac{19}{3} - \text{their } -\frac{9}{2}$ oe

9(a)	$[AB^2] = (3 - 0)^2 + (3 - 2)^2$ oe or better	M1	or $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ oe
	$[AC^2] = (0 - 2)^2 + (4 - 0)^2$ oe or better	M1	or $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$ oe
	$[BC^2] = (0 - 3)^2 + (4 - 3)^2$ oe or better	M1	or $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ oe

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10(a)(i)	$(4.5, -1)$	2	B1 for each
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				or for $y = 4$ not in coordinate form	
	10(b)(iii)	$[y =]\frac{3}{4}x + \frac{1}{2}$ final answer	3	M1 for gradient $\frac{3}{4}$ or $\frac{-1}{their(b)(i)}$ oe or better M1 for (6, 5) substituted into $y = \frac{3}{4}x + c$ or $y = their\ mx + c$ oe	

10(a)	$2 - 2x$	2	B1 for $k - 2x$ or $2 - kx$ or $3 + 2 - 2x$
10(b)(i)	Gradient (m) = correct substitution of -1 into <i>their</i> (a) $2 - 2(-1)$	M1	
	$0 = \textit{their } m \times -1 + c$ or $y [-0] = \textit{their } m(x - -1)$ oe	M1	Dep on previous M1
	$c = 4$ and leading to $y = 4x + 4$	A1	

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				B2 for $x = 1$ or M2 for $4 - (x - 1)^2$ or M1 for $(x - 1)^2$	
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